



PRAYAS

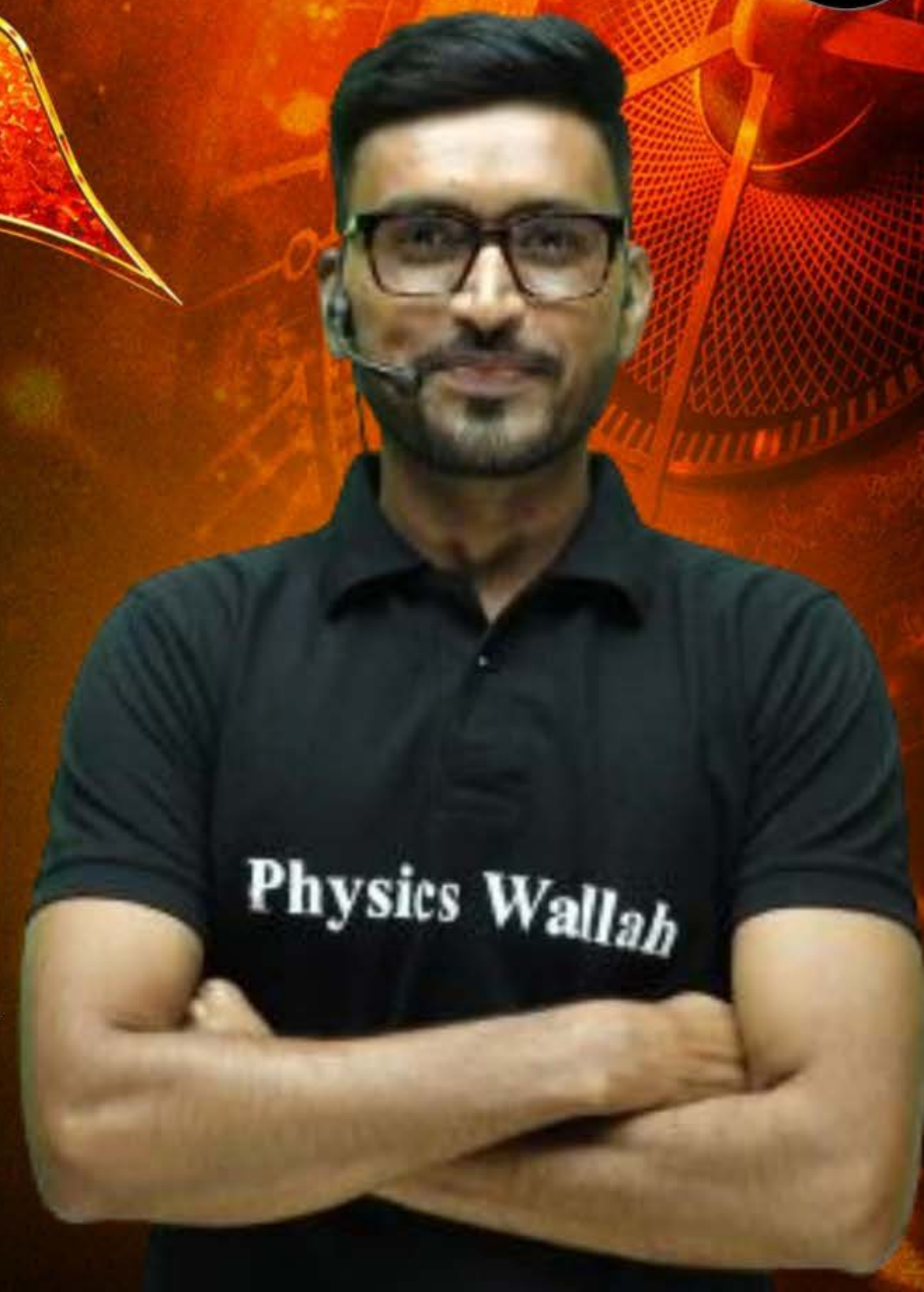
2.0 2027

Chemistry

Lecture - 13

Some Basic Concepts of Chemistry

By – Rahul Dudi Sir





Topics to be covered



- 1 Questions based on % purity
- 2 Questions based on sequential reactions
- 3 Questions based on parallel reactions
- 4 POAC, % labelling of oleum



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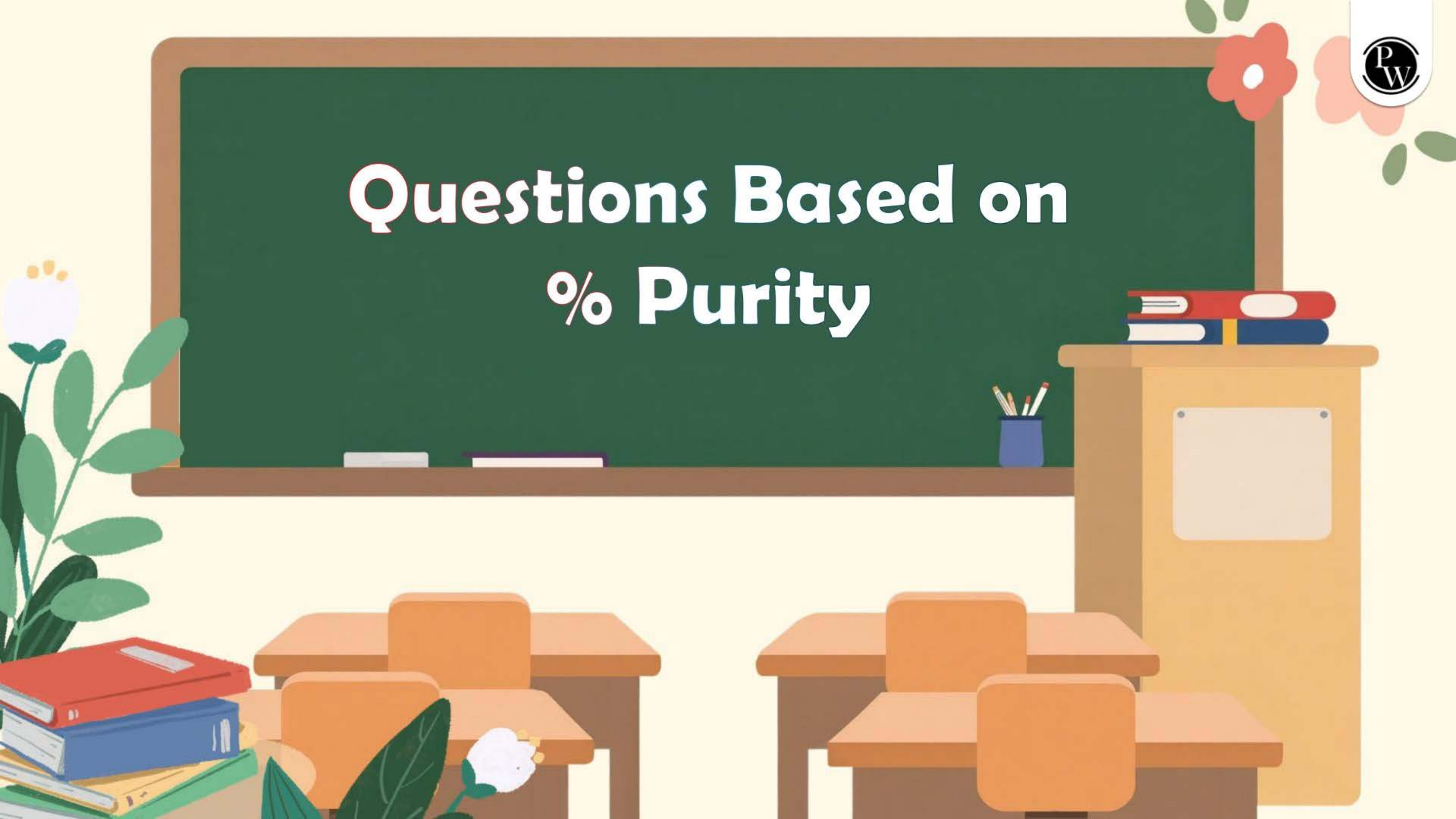
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@RAHULDUDICHEMISTRY



Questions Based on % Purity



QUESTION

①

Impure sample.

$$V_m = 22.4 \text{ L}$$

Done ✓



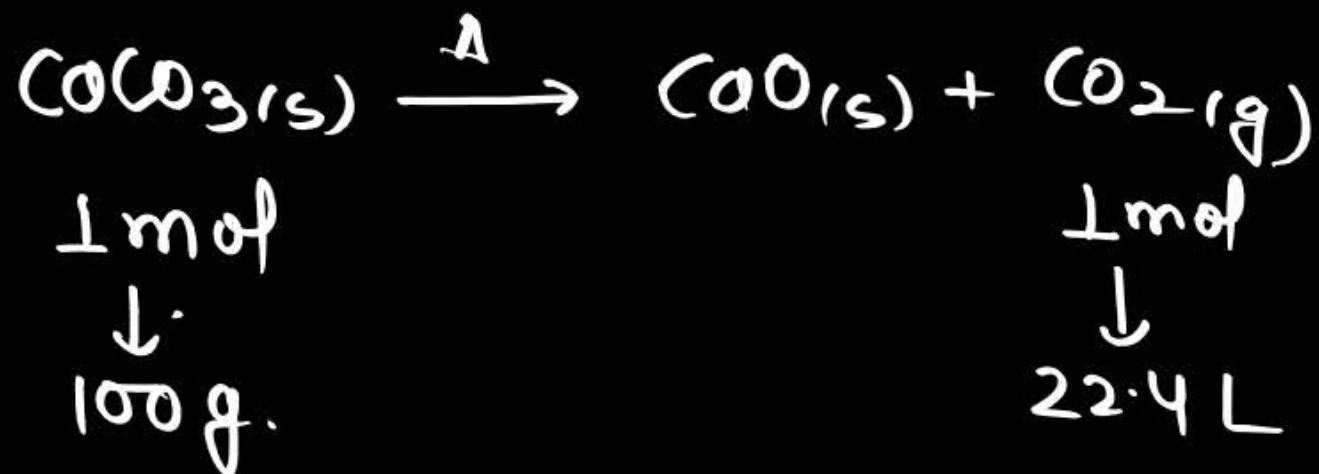
% purity of 150 g CaCO_3 sample which gives 5.6 litre of CO_2 at STP on complete decomposition is: ($P = 1 \text{ atm}$)

1 11.33 %

2 16.67 %

3 23.77 %

4 33.33 %



$$\therefore 22.4 \text{ L CO}_2 \longrightarrow 100 \text{ g pure CaCO}_3$$

$$\therefore 5.6 \text{ L CO}_2 \longrightarrow \frac{100 \times 5.6}{22.4} = 25 \text{ g pure CaCO}_3$$

$$\% \text{ Purity} = \frac{\text{Pure CaCO}_3 \times 100}{\text{Impure sample}}$$

$$= \frac{25}{150} \times 100\%$$

$$\Rightarrow 16.67\% \text{ Ans}$$

QUESTION

②

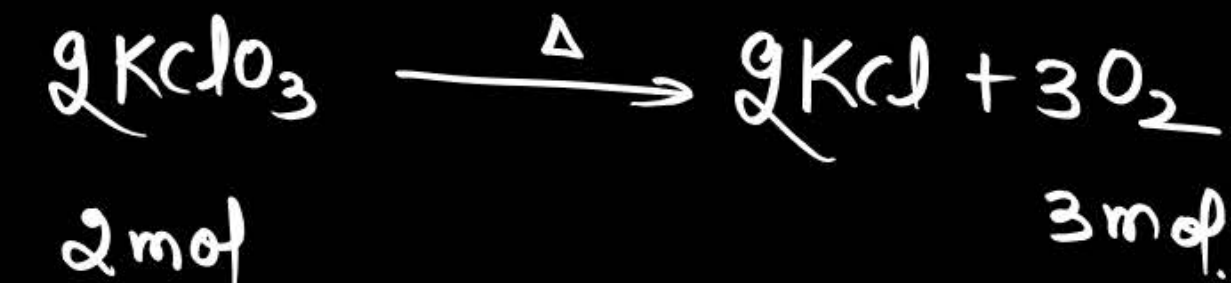


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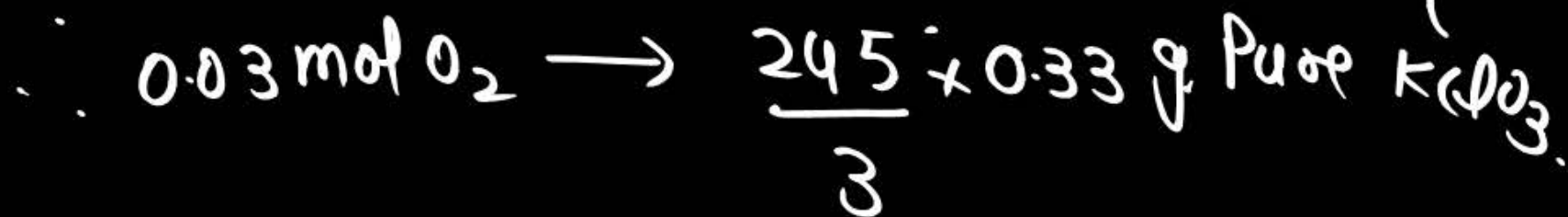
Impure sample. $MW = \frac{39}{35.5} \Rightarrow 122.5 \text{ g/mol.}$
48

5 g of KClO_3 gave 0.03 mol of O_2 . find the % purity of KClO_3 .

Ans:→



$\downarrow$
 $2 \times 122.5 \text{ g} = 245 \text{ g}$



$$\% \text{ Purity} = \frac{\text{Pure KClO}_3}{\text{Impure sample}} \times 100\%$$

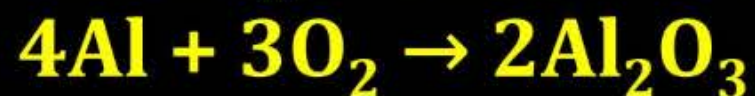
Ans:→ 49 %

QUESTION

3 HW



Based on the following consecutive reactions



2 moles of KClO_3 of 50% purity on strong heating and then with excess of Al, Al_2O_3 formed is

1 2 mol

2 1 mol

3 3 mol

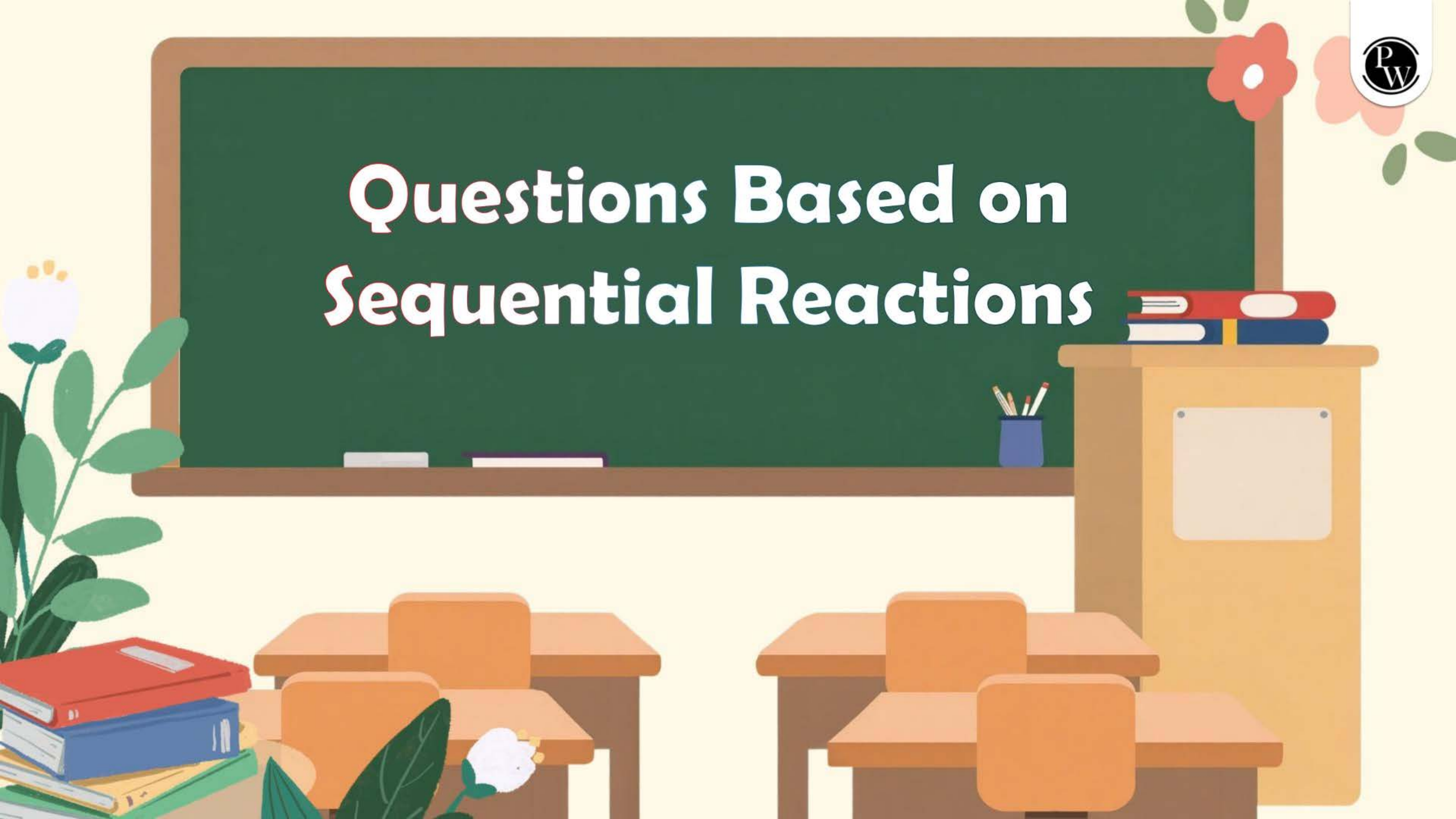
4 1.5 mol



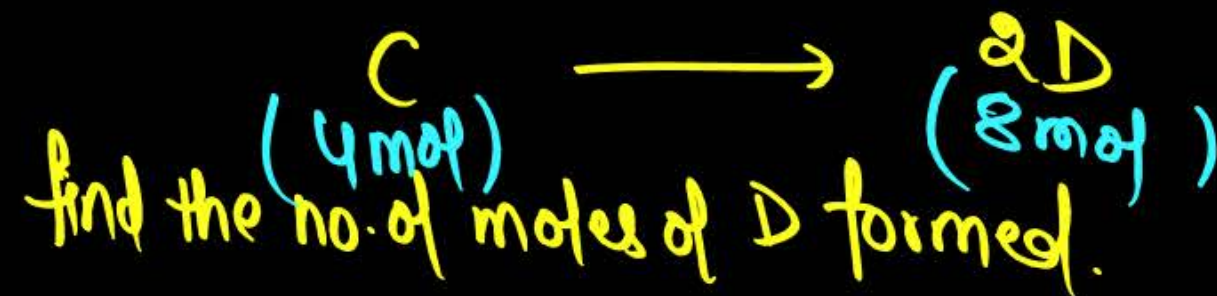
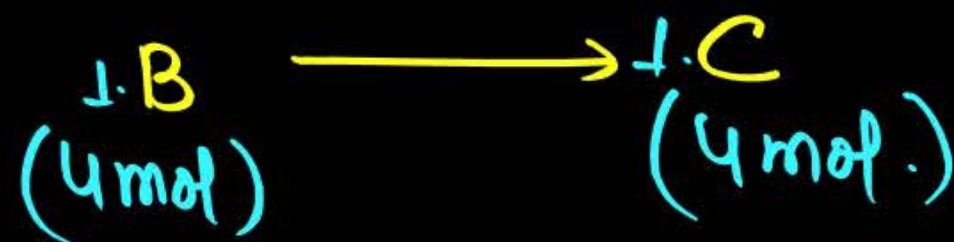
Ans. [2]



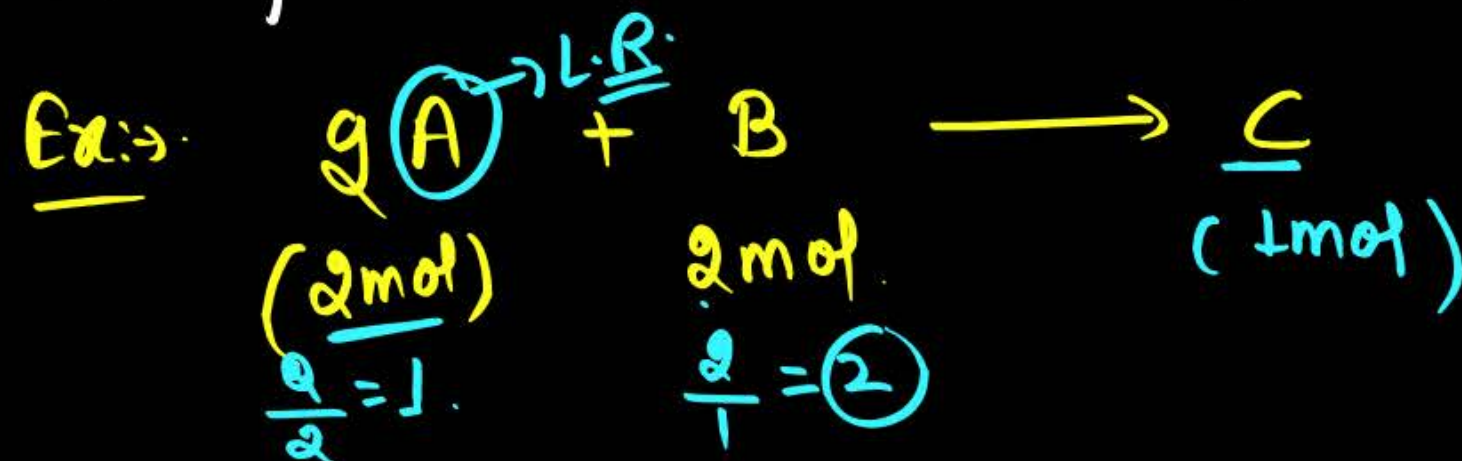
Questions Based on Sequential Reactions



* $A \longrightarrow B$
 $B \longrightarrow C$
 $C \longrightarrow A$ } sequential Rxⁿ $\Rightarrow$ Rxⁿ in which Product of one reaction is the reactant of another reaction.



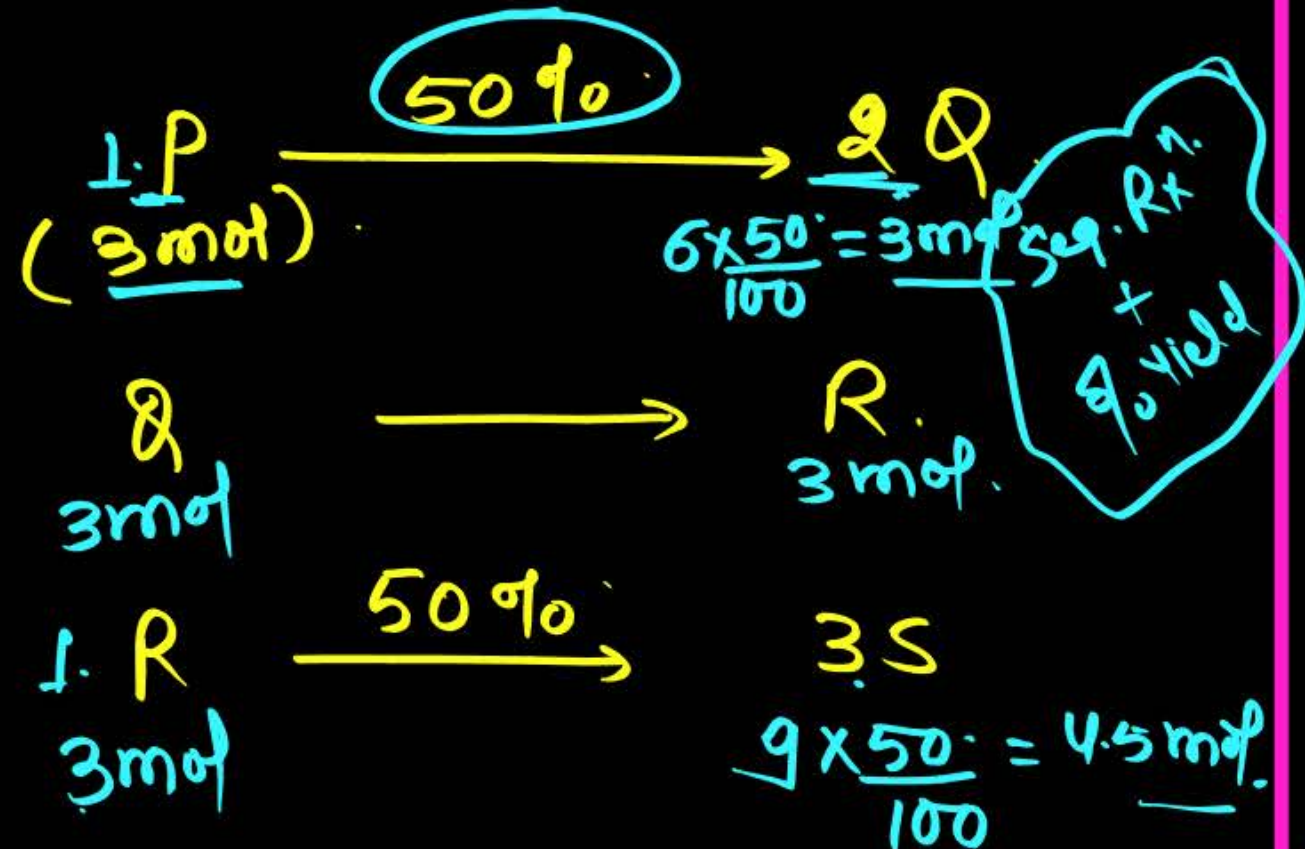
Ans: $\xrightarrow{8 \text{ mol}}$ Ans.



Ans: $\therefore 2 \text{ mol } A \rightarrow 1 \text{ mol } C$

Done

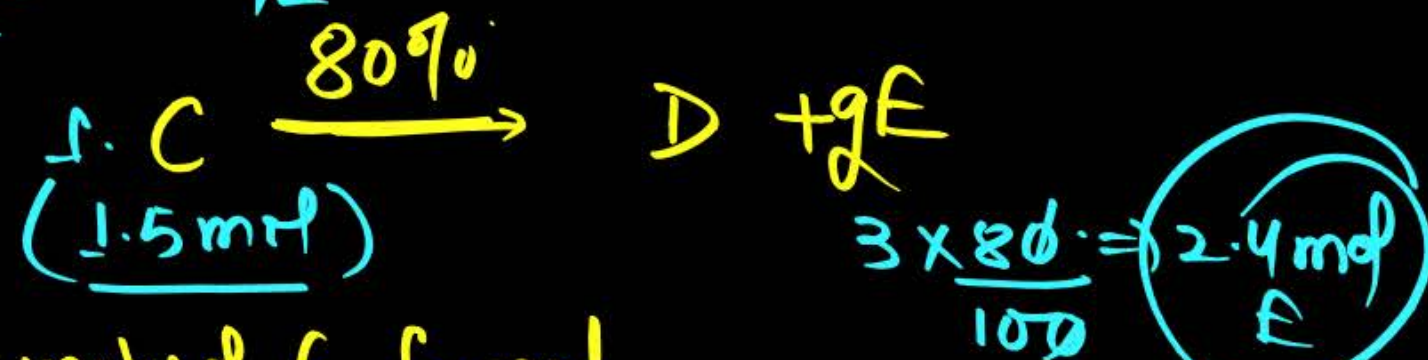
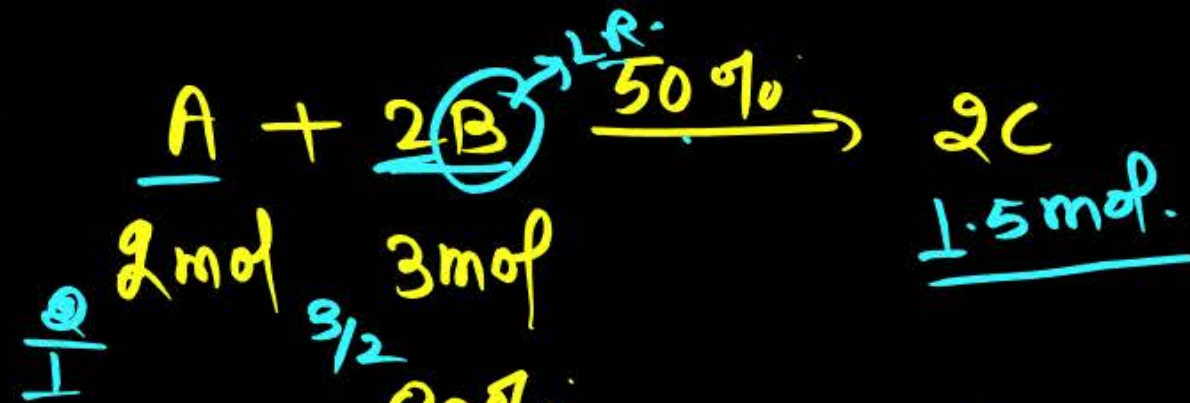
Ex: $\rightarrow$



find the moles of S formed.

Ans:-> 4.5 mol Ans.

Ex: →



find the moles of E formed

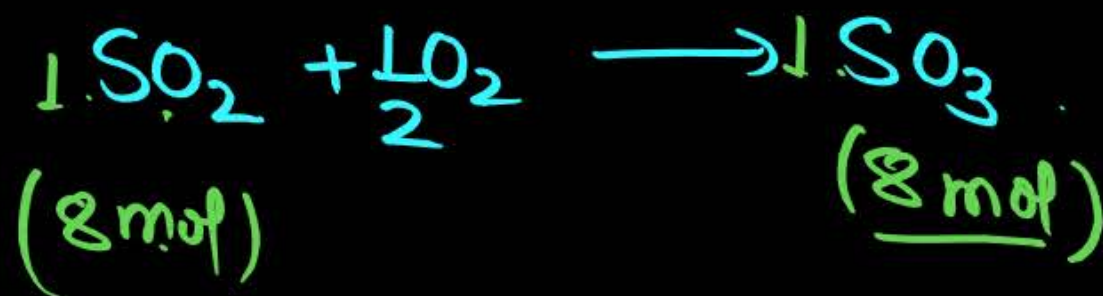
Ans:-



Sulphur trioxide is prepared by following two reactions:



How many grams of SO_3 are produced from 1 mole of S_8 ?



$$\text{Mols of SO}_3 = n_{\text{SO}_3} \times \text{MW}$$

$$\Rightarrow 8 \times 80$$

$$\Rightarrow \underline{640 \text{ g.}} \text{ Ans.}$$

QUESTION

5

K-39, Cl-35.5, O-16.

Done



KClO₄ can be prepared by Cl₂ and KOH by a series of reactions as given below



(i) Calculate mass of Cl₂ in gram required to produce 1385 gm KClO₄?

$$n_{\text{KClO}_4} = \frac{w}{M_w} = \frac{1385}{138.5} \Rightarrow 10 \text{ mol}$$

$$M_w = \begin{array}{r} 39 \\ 35.5 \\ 64 \\ \hline 138.5 \text{ g/mol} \end{array}$$

Ans: $\therefore 3 \text{ mol KClO}_4 \longrightarrow 4 \text{ mol KClO}_3$
 $\therefore 10 \text{ mol KClO}_4 \longrightarrow \frac{4 \times 10 \text{ mol KClO}_3}{3} \Rightarrow \frac{40}{3} \text{ mol}$

$\therefore 1 \text{ mol KClO}_3 \longrightarrow 3 \text{ mol KClO}$
 $\therefore \frac{40}{3} \text{ mol KClO}_3 \rightarrow 3 \times \frac{40}{3} \Rightarrow 40 \text{ mol KClO}$

moles of Cl₂ required = 40

Mass of Cl₂ req. $\Rightarrow n_{\text{Cl}_2} \times M_w$

$\Rightarrow 40 \times 71$

$\Rightarrow \underline{2840 \text{ g.}} \quad \text{Ans}$

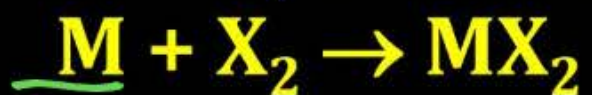
QUESTION

6

HW



NX is produced by the following step of reactions:



How much M (metal) is consumed to produce 206 gm of NX?

(Take at. Wt. of M = 56 N = 23 x = 80)

1 42 gm

2 56 gm

3 14/3 gm

4 7/4 gm



Ans. [1]

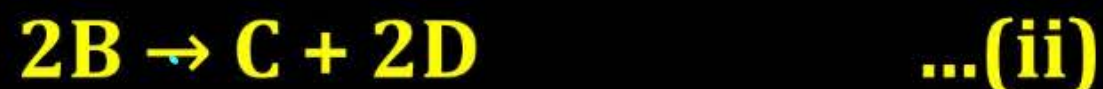
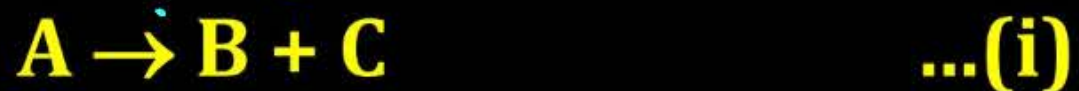
QUESTION

7

Hw



For sequential reaction:



If % yield of (i) and (ii) reactions are 90% and 80% respectively then the overall % yield is expected to be:

1 90%

2 80%

3 72%

4 10%



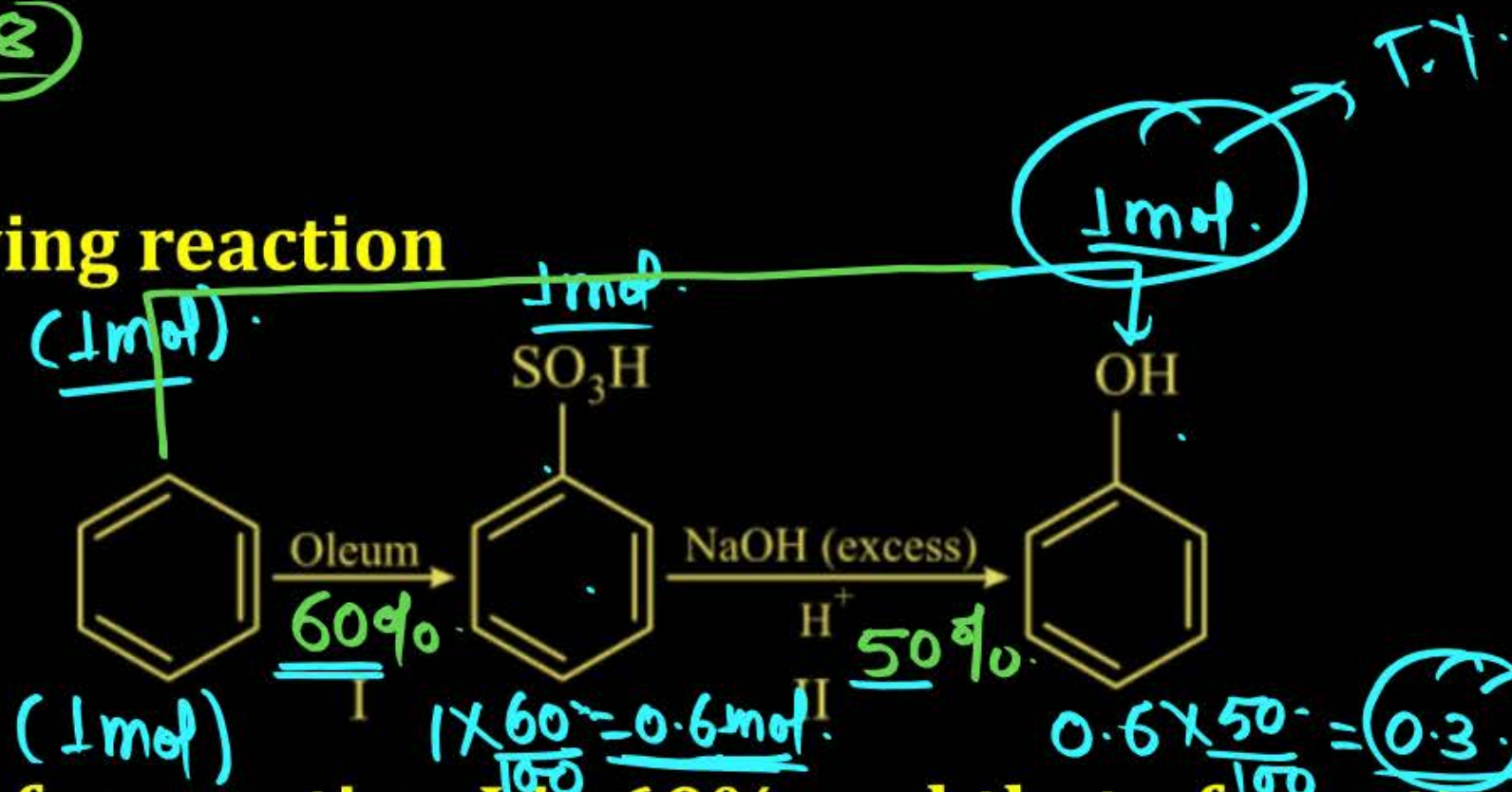
Ans. [3]

QUESTION

8



In the following reaction



$$\begin{aligned} \% \text{ Yield} &= \frac{\text{A.Y.}}{\text{T.Y.}} \times 100\% \\ &= \frac{0.3}{1} \times 100\% \\ &= 30\% \end{aligned}$$

The % yield for reaction I is 60% and that of reaction II is 50%. The overall yield of the complete reaction is _____% [Nearest integer] [27 July, 2022 (Shift-I)]

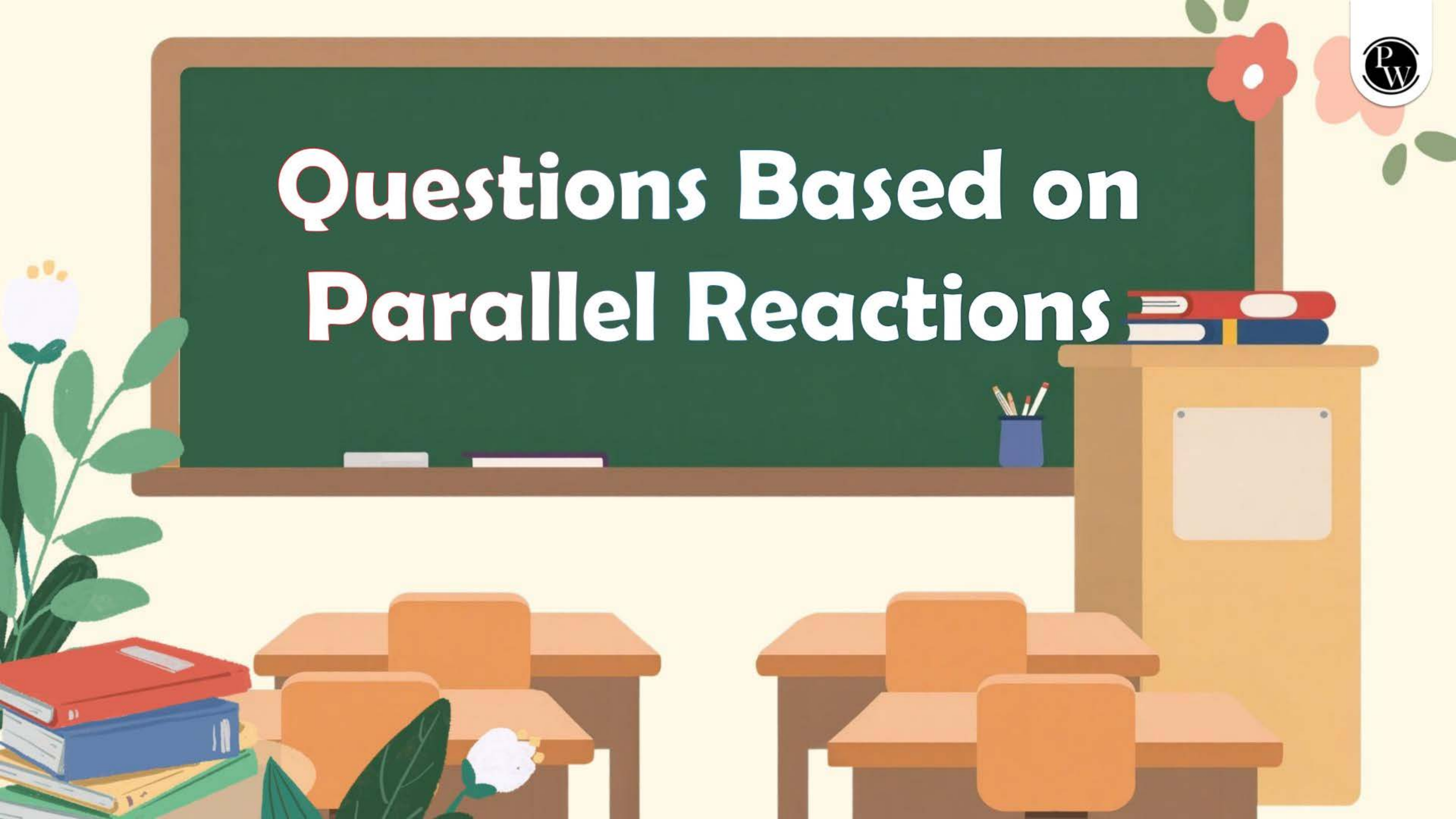
Ans:→

Done

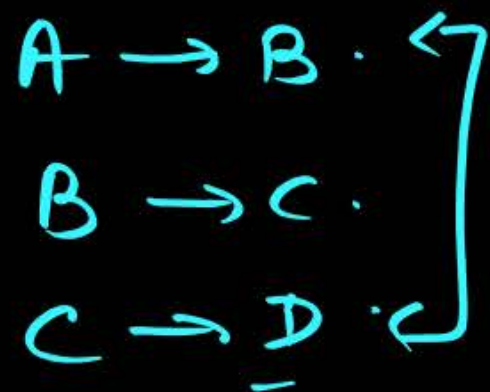
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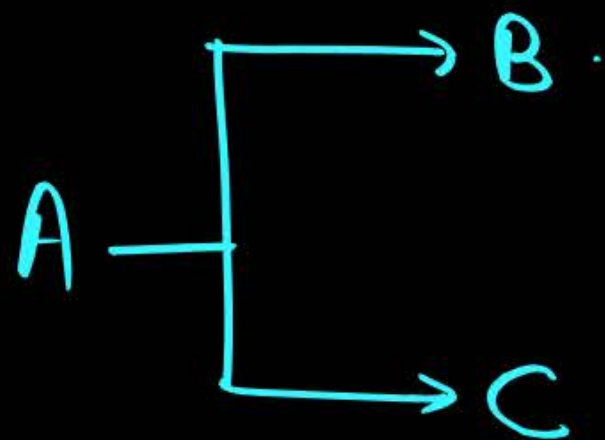
Questions Based on Parallel Reactions



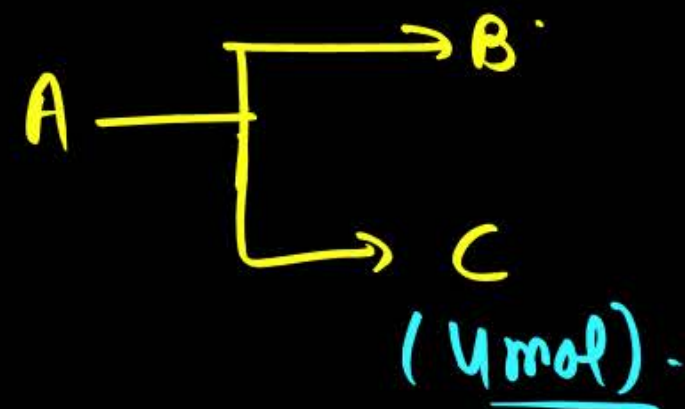
* Sequential Rxn $\Rightarrow$



* Parallel Rxn $\Rightarrow$

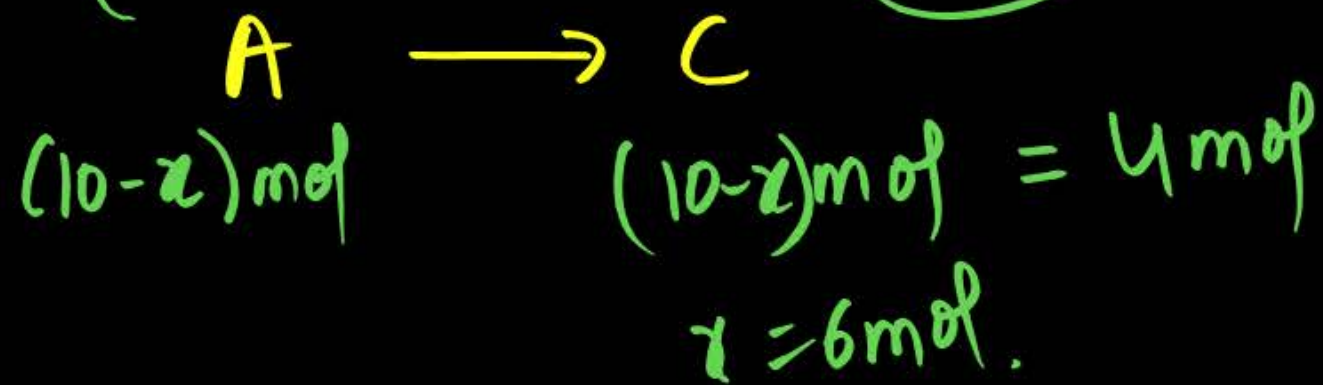


Ex $\Rightarrow$



If 10 moles of A are taken initially. find the no. of moles of B formed.

Ans $\Rightarrow$

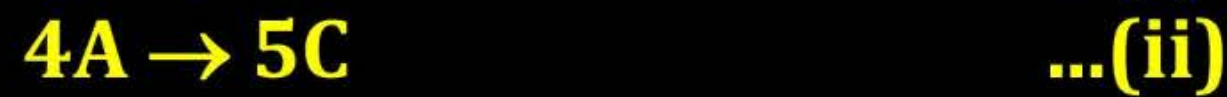
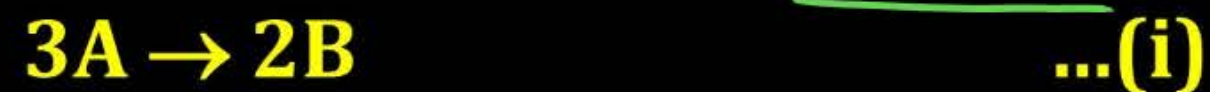


QUESTION

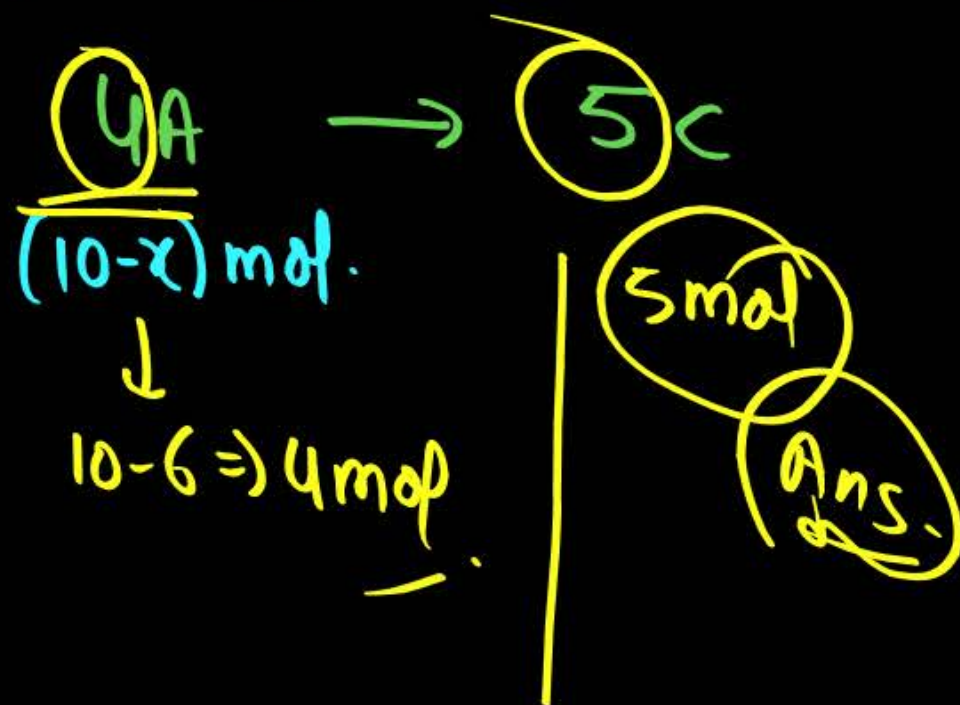
9



10 moles of A are taken in a closed container. If 4 moles of B are formed then calculate moles of C formed.



Ans:→



5 mol
Ans.



$$\Rightarrow \underline{x = 6 \text{ mol.}}$$

If 6 moles of KClO_3 are decomposed according to following reactions calculate the moles of KClO_4 , produced if mole of O_2 produced are 3?



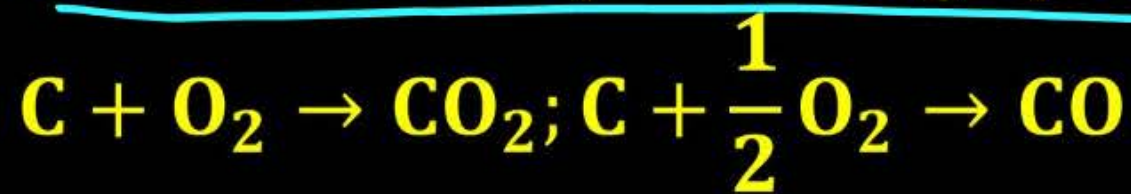


Ans. [3]

QUESTION



If 240 g of carbon is taken in a container to convert it completely to CO_2 but in industry it has been found that 280 g of CO was also formed along with CO_2 . Find the mole percentage yield of CO_2 . The reactions occurring are:



1 25

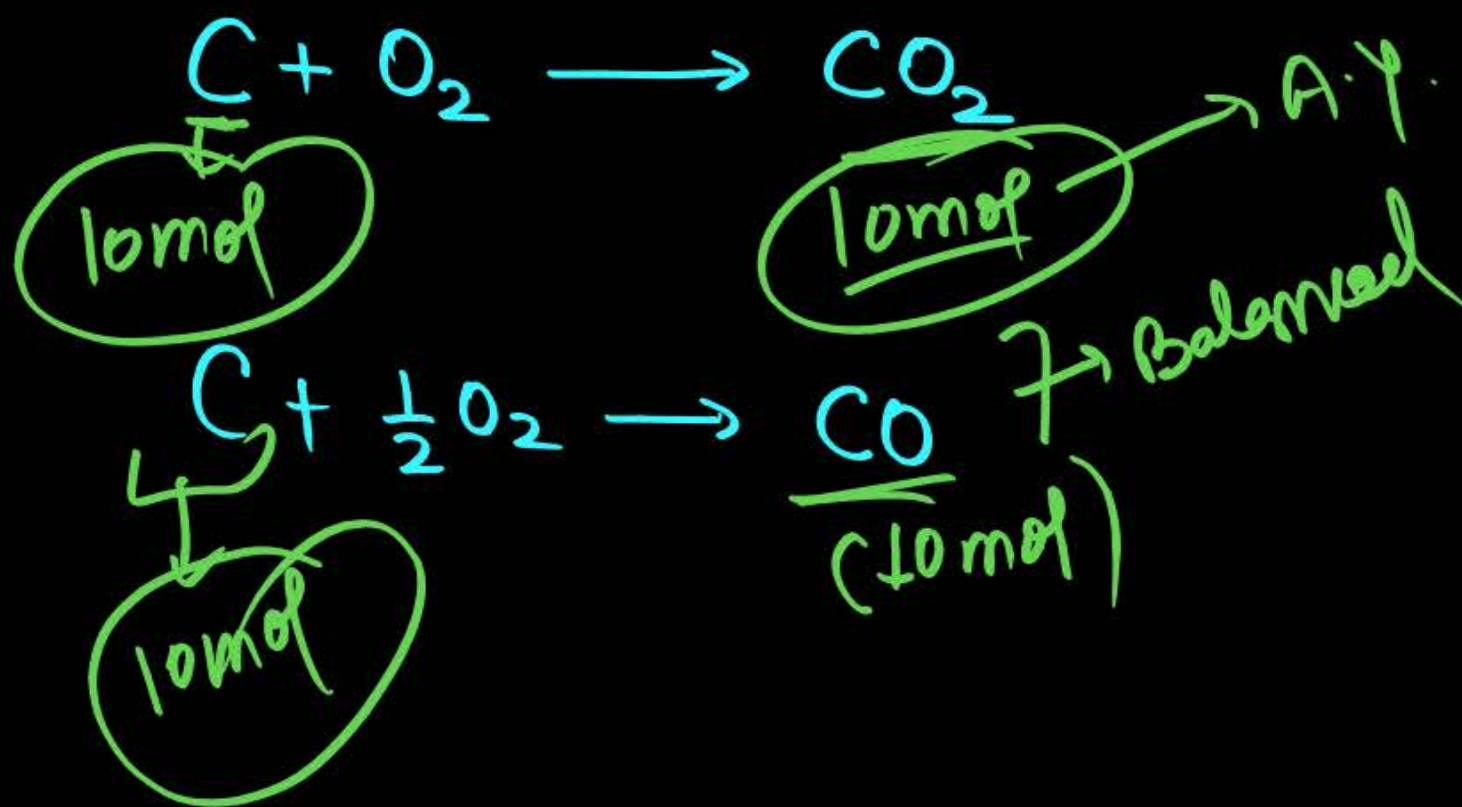
2 50

3 75

4 35

$$n_{\text{C}} = \frac{240}{12} = 20 \text{ mol}$$

$$n_{\text{CO}} = \frac{w}{m_w} = \frac{280}{28} = 10 \text{ mol}$$



$$\% \text{ yield} = \frac{\text{A.Y.}}{\text{T.Y.}} \times 100\%$$

$$\Rightarrow \frac{10}{20} \times 100\%$$

$$\Rightarrow 50\% \text{ (Ans)}$$

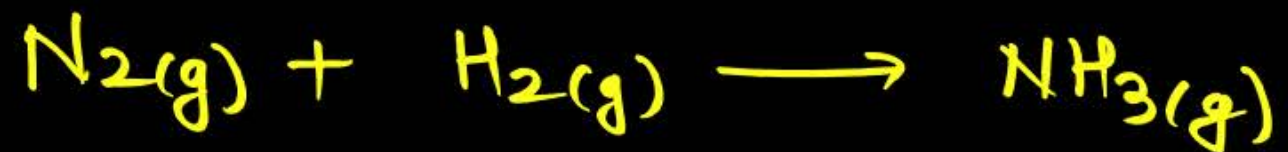


Two substances A_2 , and B_2 , are allowed to react completely to form A_2B_4 and A_3B_4 mixture, leaving none of the reactions. Using this information, Calculate the composition of final mixture when $5/4$ mole of A_2 and 2 mole of B_2 is taken?



Ans. $[A_3B_4 = 0.5 \text{ mole}, A_2B_4 = 0.5 \text{ mole}]$

* POAC $\Rightarrow$



POAC for N-atoms

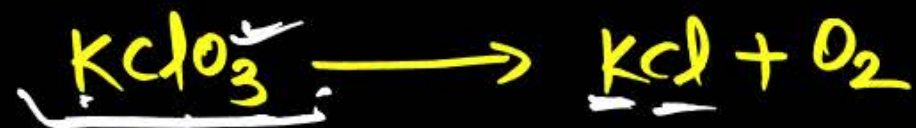
moles of N-atoms on reactant side = moles of N-atoms on product side

$$2 \times \text{moles of N}_2 = 1 \times \text{moles of NH}_3$$

POAC for H-atoms $\Rightarrow$

moles of H-atoms on reactant side = moles of H-atoms on product side

$$2 \times \text{moles of H}_2 = 3 \times \text{moles of NH}_3$$



Done



POAC for K-atoms $\Rightarrow$ moles of K-atoms on reactant side = moles of K-atoms on product side.

$$1 \times \text{moles of KClO}_3 = 1 \times \text{moles of KCl.}$$

POAC for Cl-atoms $\Rightarrow$

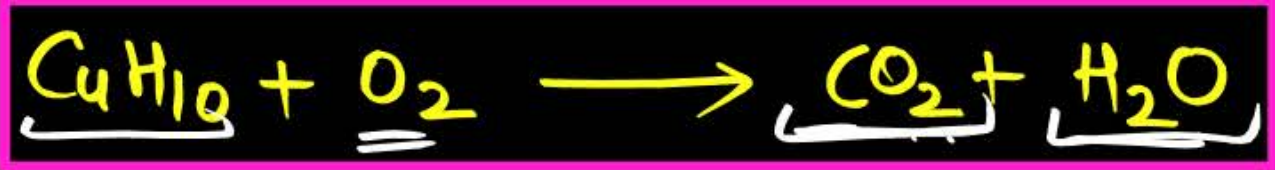
moles of Cl-atoms on reactant side = moles of Cl-atoms on product side

$$1 \times \text{moles of KClO}_3 = 1 \times \text{moles of KCl.}$$

POAC for O-atoms $\Rightarrow$

moles of O-atoms on reactant side = moles of O-atoms on product side.

$$3 \times \underline{\eta_{\text{KClO}_3}} = 2 \times \eta_{\text{O}_2}$$



* POAC for C-atoms:-

moles of C-atoms on reactant side = moles of C-atoms on product side

$$4 \times n_{\text{C}_4\text{H}_{10}} = 1 \times n_{\text{CO}_2}$$

* POAC for H-atoms:-

moles of H-atoms on reactant side = moles of H-atoms on product side

$$10 \times n_{\text{C}_4\text{H}_{10}} = 2 \times n_{\text{H}_2\text{O}}$$

* POAC for O-atoms:-

moles of O-atoms on reactant side = moles of O-atoms on product side

$$2 \times n_{\text{O}_2} = 2 \times n_{\text{CO}_2} + 1 \times n_{\text{H}_2\text{O}}$$



Principle of Atom Conservation (POAC)

- The Principle of Atom Conservation (POAC) states that the total number of atoms of reactants must equal the number of atoms of products.
- Number of atoms of element in reactant = Number of atoms of element in product
- Moles of atoms of element in reactant = moles of atoms of element in product

QUESTION

13



Find out the moles of CO_2 formed from combustion of 60 moles of butane.

Ans: $\rightarrow$ $n \neq 1$



$\downarrow$
1 mol.

$\downarrow$
4 mol.

60 mol $\longrightarrow$ 240 mol Ans.

$n \neq 2$



POAC for C-atoms:

$$4 \times n_{\text{C}_4\text{H}_{10}} = 1 \times n_{\text{CO}_2}$$

$$4 \times 60 = n_{\text{CO}_2}$$

$$\underline{240 = n_{\text{CO}_2}} \text{ Ans.}$$

QUESTION

HW 14



What mass of O_2 is produced by 1.23 g of $KClO_3$?

3.37 g PtCl_4 has gone through a number of chemical process treatments and finally all of chlorine is converted to a complex $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{SO}_4$. The final weight of the product formed is:

[Molar mass of $\text{PtCl}_4 = 337 \text{ g mol}^{-1}$, Molar mass of $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{SO}_4 = 482 \text{ g mol}^{-1}$]



* POAC for Cl-atoms $\Rightarrow$

$$4 \times n_{\text{PtCl}_4} = 2 \times n_{[\text{Pt}(\text{en})_2\text{Cl}_2]\text{SO}_4}$$

$$4 \times \left(\frac{3.37}{337} \right) = 2 \times \frac{\text{mass}}{482}$$

$$\frac{241}{100}$$

$$\text{mass} = \frac{4 \times 241}{100}$$

$$\Rightarrow \underline{9.64 \text{ g.}}$$

QUESTION

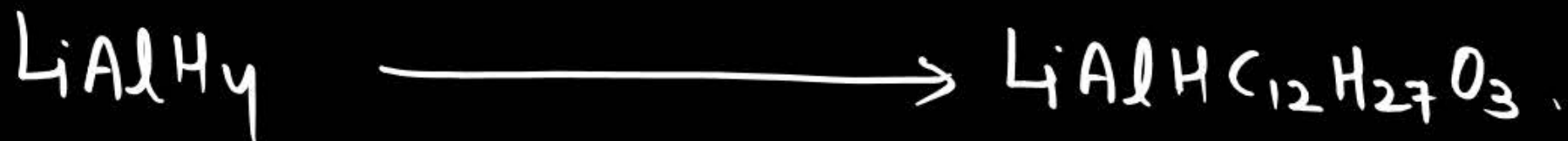
16



0.32 mole of LiAlH_4 in ether solution was placed in a flask and 74 gm (1 mole) of t-butyl alcohol was added. The product is $\text{LiAlHC}_{12}\text{H}_{27}\text{O}_3$. Calculate mass of product if all lithium atoms are converted. $\text{Li}-7, \text{Al}-27, \text{C}-12, \text{H}-1, \text{O}-16$

$$M_w = 254 \text{ g/mol}$$

Ans:→



POAC for Li-atoms:→

$$1 \times n_{\text{LiAlH}_4} = 1 \times n_{\text{product}}$$

$$0.32 = \frac{\text{mass}}{254}$$

$$\text{mass} = (254 \times 0.32) \text{ g.}$$

$$= \underline{81.28 \text{ g. Ans}}$$

27.6 gm of K_2CO_3 was treated by a series of reagents so as to convert all of its carbon to $K_2Zn_3[Fe(CN)_6]_2$. Calculate the weight of the product. [mol. wt. of $K_2CO_3 = 138$ and that of $K_2Zn_3[Fe(CN)_6]_2$ is 698]



POAC for C-atoms:

$$1 \times n_{K_2CO_3} = 12 \times n_{\text{Product}}$$

$$\frac{27.6}{138} = 12 \times \frac{\text{mass}}{698}$$

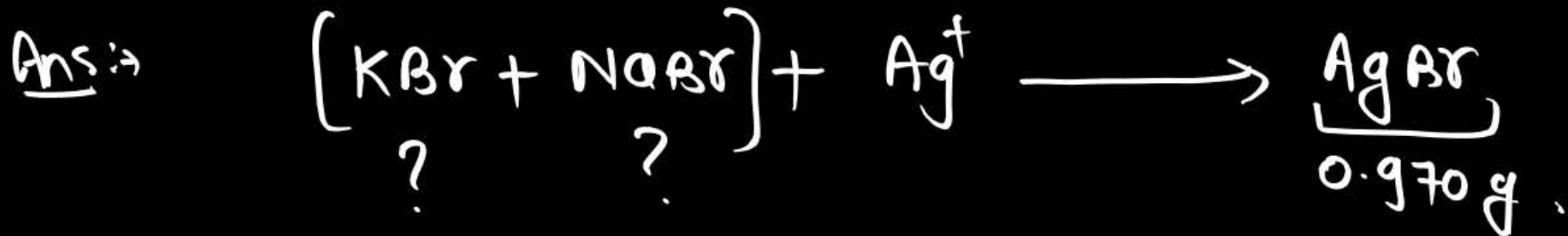
$$\text{Mass} = \text{g. } \underline{11.634}$$

QUESTION



A mixture of KBr and NaBr weighing 0.560 gm was treated with aq. Ag^+ ion and all bromide ion was recovered as 0.970 gm of pure AgBr. What was the fraction by weight of KBr in the sample?

[Ag - 108, I - 127, Br - 80]



POAC for Br-atoms:-



Percentage Labelling of Oleum

- Oleum is fuming sulphuric acid which contains extra SO_3 dissolved in H_2SO_4 .
- To convert this extra SO_3 into H_2SO_4 , water has to be added.
- The amount of sulphuric acid obtained when just sufficient water is added into 100 g of oleum so that all SO_3 present in it is converted into H_2SO_4 , is called percentage labelling of oleum.

* oleum $\therefore \rightarrow$.

fuming H_2SO_4 .



10g % oleum.

% labelling of oleum.

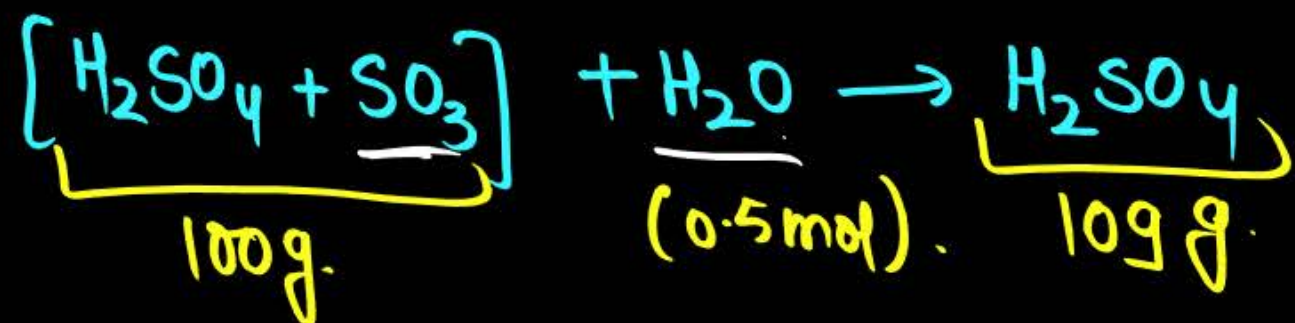
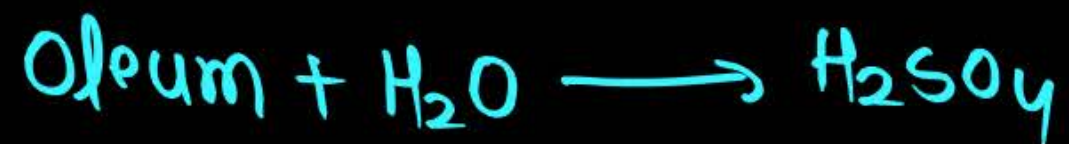


Percentage Labelling of Oleum



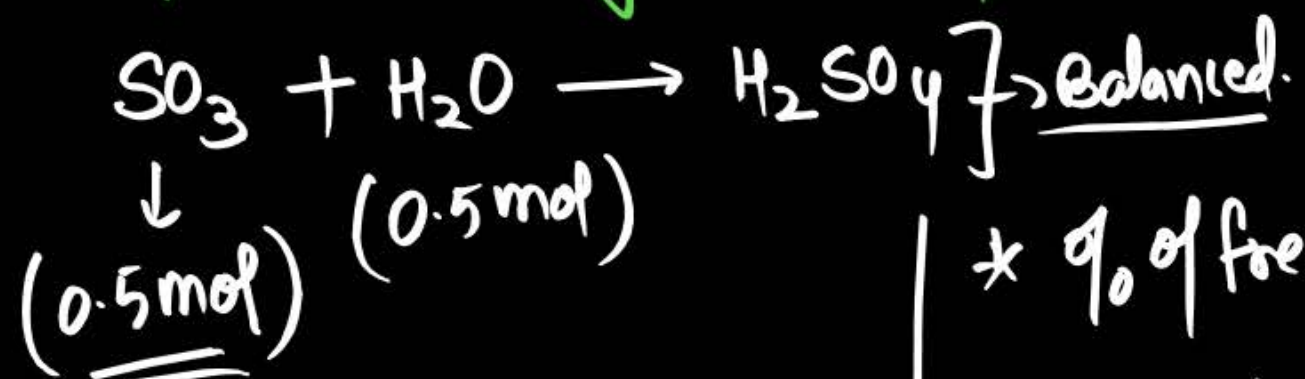
Example: 109 % Oleum

100 g. oleum reacts with water to form 109 g. of H_2SO_4 .



$$\text{Mass of H}_2\text{O} = 109 - 100 = 9\text{g.}$$

$$n_{\text{H}_2\text{O}} = \frac{9}{18} = 0.5\text{mol.}$$



$$\begin{aligned} \text{Mass of SO}_3 &= n_{\text{SO}_3} \times \text{Mw} \\ &= 0.5 \times 80 \\ &= \underline{40\text{g.}} \end{aligned}$$

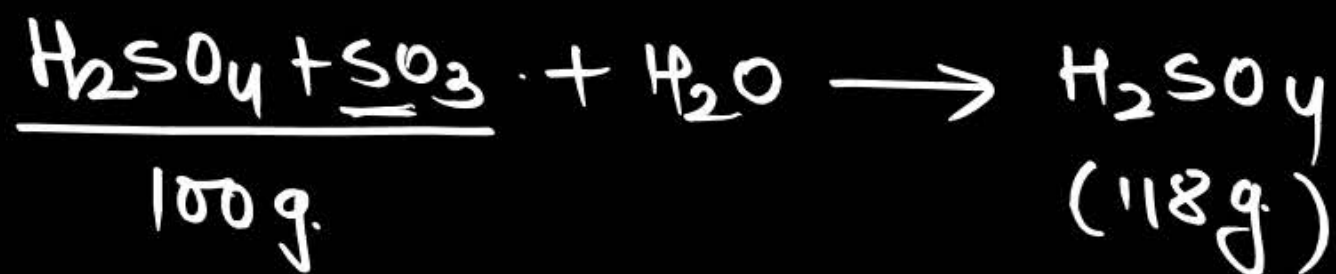
* % of free SO_3 in oleum

$$\Rightarrow \frac{40}{100} \times 100\%$$

$$\Rightarrow \underline{40\%} \quad (\text{Ans})$$

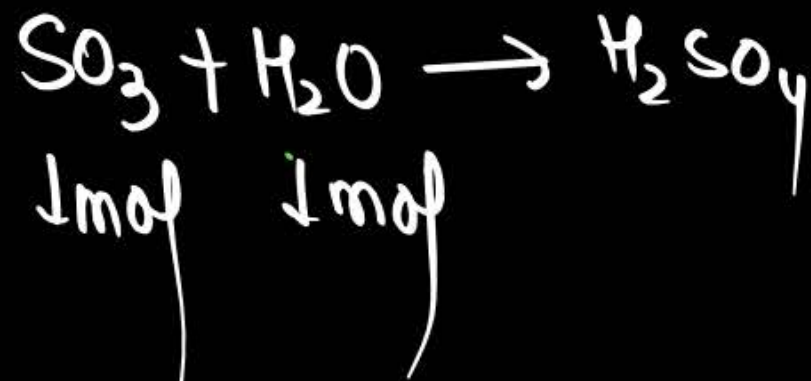
Calculate the % of free SO_3 in an oleum, that is labelled 118%.

Ans: $\Rightarrow$ 118% oleum $\Rightarrow$ 100 g. oleum reacts with H_2O to form 118 g. H_2SO_4 .



Mass of $\text{H}_2\text{O} \Rightarrow 118 - 100 = 18 \text{ g.}$

$$\eta_{\text{H}_2\text{O}} = \frac{18}{18} \Rightarrow \textcircled{1}$$



$$\begin{aligned} \text{Mass of SO}_3 &= n_{\text{SO}_3} \times \text{MW} \\ &\Rightarrow 1 \times 80 = \underline{80 \text{ g.}} \end{aligned}$$

$$\begin{aligned} \% \text{ free SO}_3 \text{ in oleum} &= \frac{80}{100} \times 100 \% \\ &= \underline{80 \%} \cdot \textcircled{\text{Ans}} \end{aligned}$$

$$\% \text{ of free } \text{SO}_3 \text{ in oleum} = \frac{80(y-100)}{18} \cdot \%.$$

% labelling of oleum.

Which of the following percentage strength is not possible for a sample of oleum?

1 104 %

2 127 %

3 118 %

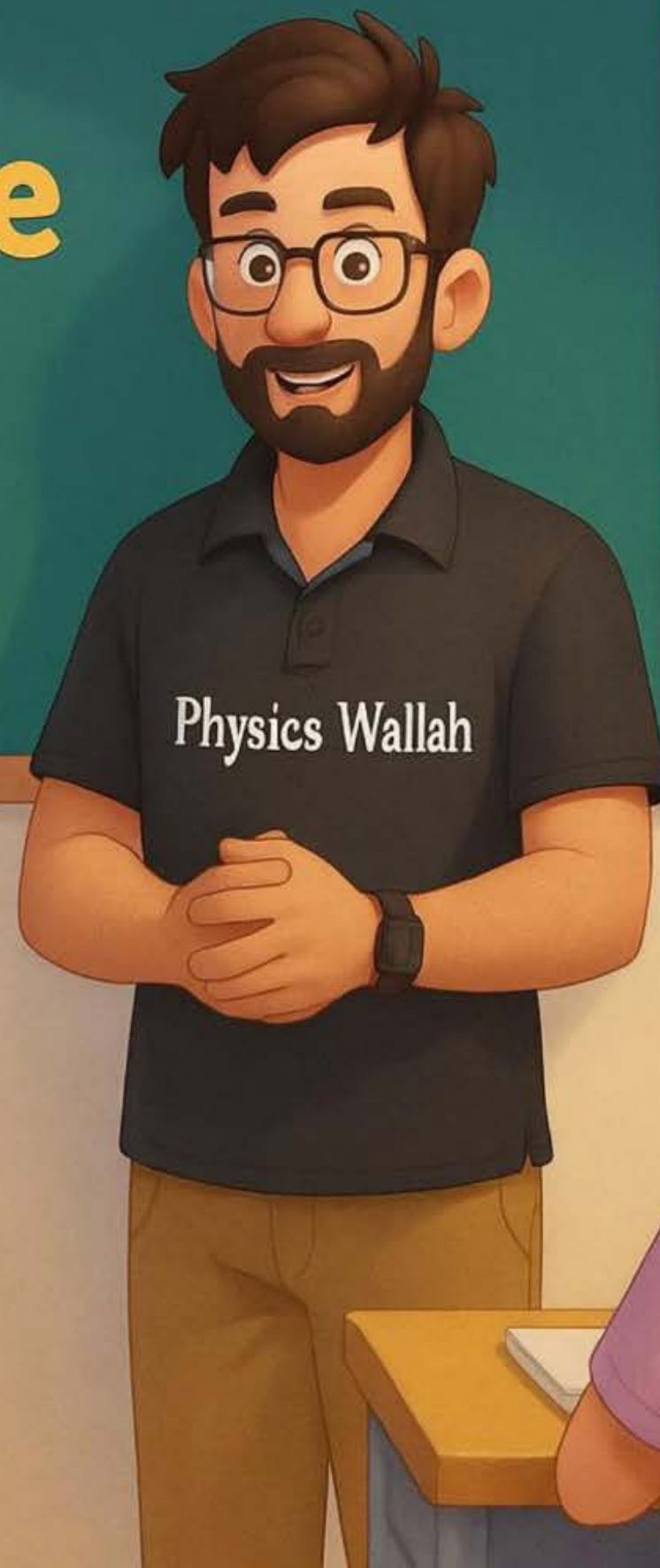
4 109 %

$$\begin{aligned}\% \text{ of free SO}_3 &= \frac{80}{18} (y - 100) \\ &= \frac{80}{18} \times (127 - 100) \\ &= \frac{80}{18} \times \frac{40}{2} \times 3 \\ &\Rightarrow 120\% \end{aligned}$$

If in a sample of oleum, mole fraction of SO_3 is 0.5. Label the oleum sample.



**Aaj ke liye
itna hi**





Homework

➤ Examples:

46- 51

➤ Check your understanding:

79-85

➤ Topicwise :

29-36

Complete -

* Normality
law of eq.
eq. weight
nf. } X

* Vol. strength of H_2O_2 . } X

Thank

You